Lecture 17, Math 441

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Constant Coefficient n_{th} Order Linear Homogeneous Equations

Ex: $y''' - 3y'' + 4y' = 0$

$y(x) = \mathrm{e}^{rx}$ and get the characteristic equation:

$r^3 - 3r^2 + 4r = r(r^2 - 3r + 4) = 0$

$r = \frac{3 \pm \mathrm{i}\sqrt{7}}{2} \qquad r = 0$

$y_1 = \mathrm{e}^{0x} = 1,\ y_2 = \mathrm{e}^{\frac{3+\mathrm{i}\sqrt{7}}{2}x},\ y_3 = \mathrm{e}^{\frac{3-\mathrm{i}\sqrt{7}}{2}x}$

Using Euler's formulas:

$y(x) = c_1 + \mathrm{e}^{\frac{3x}{2}}\left[c_2 \cos\left(\frac{\sqrt{7}}{2}x\right) + c_3 \sin\left(\frac{\sqrt{7}}{2}x\right)\right]$

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Operator Notation

Ex: $y''' - y' = 0$. If we use D for the derivative operator, $y' = \mathrm{D}y$,

we can rewrite the equation in operator notation:

$$\mathrm{D}^3 y - \mathrm{D}y = (\mathrm{D}^3 - \mathrm{D})y = 0$$

$$\mathrm{D}(\mathrm{D} - 1)(\mathrm{D} + 1)y = 0$$

$$\mathrm{D}(\mathrm{D} + 1)(\mathrm{D} - 1)y = 0$$

$$\mathrm{D} + 1))(\mathrm{D} - 1)\mathrm{D}y = 0$$

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$$(\mathrm{D}+1)y = 0.$$

$$\mathrm{D}y + y = 0$$

This is $y' = -y \to y = \mathrm{e}^{-x}$

$$(\mathrm{D}-1)y = 0 \to y = \mathrm{e}^{x}$$

$$\mathrm{D}y = 0 \to y = 1$$

$$y = c_1 1 + c_2 \mathrm{e}^{x} + c_3 \mathrm{e}^{-x}$$

Repeated Roots

Ex: $(\mathrm{D}-2)^4 y = 0$

$(\mathrm{D}-2)y = 0 \rightarrow y = \mathrm{e}^{2x}$

A single factor $(\mathrm{D}-2)y = 0$ has the solution $y = \mathrm{e}^{2x}$

$y_1 = \mathrm{e}^{2x} \qquad y_2 = x\mathrm{e}^{2x} \qquad y_3 = x^2\mathrm{e}^{2x} \qquad y_4 = x^3\mathrm{e}^{2x}$

A general solution can be written as

$y(x) = c_1\,\mathrm{e}^{2x} + c_2\,x\,\mathrm{e}^{2x} + c_3\,x^2\,\mathrm{e}^{2x} + c_4\,x^3\,\mathrm{e}^{2x}$

Ex: $(\mathrm{D}^2 + 2\mathrm{D} + 5)^2 y = 0$

Solve one factor: $(\mathrm{D}^2 + 2\mathrm{D} + 5)y = 0$.

$y = \mathrm{e}^{rx}$, we get $r^2 + 2r + 5 = 0 \;\rightarrow\; r = -1 \pm 2\mathrm{i}$ (twice)

$y(x) = \mathrm{e}^{-x}\left[A\cos(2x) + B\sin(2x)\right] + x\,\mathrm{e}^{-x}\left[C\cos(2x) + D\sin(2x)\right]$

n_{th} Order Linear Homogeneous IVP

Ex: $y''' - y' = 0; \qquad y(0) = 2,\ y'(0) = 0,\ y''(0) = 2$

$y = \mathrm{e}^{rx}$ and get $r^3 - r = r(r^2 - 1) = r(r-1)(r+1) = 0$

$r = 0, r = 1, r = -1$ and a general solution is

$$\boxed{y(x) = c_1 + c_2\,\mathrm{e}^{x} + c_3\,\mathrm{e}^{-x}}$$

$y' = c_2\,\mathrm{e}^{x} - c_3\,\mathrm{e}^{-x};\ \ y'' = c_2\,\mathrm{e}^{x} + c_3\,\mathrm{e}^{-x}$

Applying the initial conditions:

$$y(0)+c_1+c_2+c_3=2; \qquad y'(0)=c_2-c_3=0; \qquad y''(0)=c_2+c_3=2$$

$$\boxed{c_2=c_3=1;\ \ c_1=0}$$

So the solution to the IVP:

$$\boxed{y(x)=\mathrm{e}^{x}+\mathrm{e}^{-x}}$$

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n_{th} Order Linear Inhomogeneous IVP

Ex: $y''' - y' = 6\,\mathrm{e}^{2x}$; $\qquad y(0) = 2,\ y'(0) = 0,\ y''(0) = 2$

$y = A\,\mathrm{e}^{2x}$

$y' = 2\,A\,\mathrm{e}^{2x}$ and $y''' = 8\,A\,\mathrm{e}^{2x}$.

Substituting, we get:

$8\,A\,\mathrm{e}^{2x} - 2\,A\,\mathrm{e}^{2x} = 6\,\mathrm{e}^{2x} \ \rightarrow 8A - 2A = 6 \rightarrow A = 1$

So a particular solution is $\boxed{y_p = \mathrm{e}^{2x}}$

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But y_p does not satisfy the initial conditions. Build a solution using

y_p and add to it: $y = y_p + y_c$ where y_c is a complementary solution.

$y' = 2e^{2x} + yc'$ $\qquad$ $y''' = 8e^{2x} + yc'''$

$8e^{2x} + yc''' - 2e^{2x} - yc' = 6e^{2x}$

$y_c''' - y_c' = 0$ $\qquad$ associated homogeneous eq. of the original eq.

$$\boxed{y_c(x) = c_1 + c_2\,\mathrm{e}^{x} + c_3\,\mathrm{e}^{-x}}$$ complementary solution

$$y = y_c + y_p = \underbrace{c_1 + c_2\mathrm{e}^{x} + c_3\mathrm{e}^{-x}}_{y_c} + \underbrace{\mathrm{e}^{2x}}_{y_p}$$ general solution

$y' = c_2\,\mathrm{e}^{x} - c_3\,\mathrm{e}^{-x} + 2\,\mathrm{e}^{2x}$; $y'' = c_2\,\mathrm{e}^{x} + c_3\,\mathrm{e}^{-x} + 4\,\mathrm{e}^{2x}$

Apply the initial conditions:

$y(0) = c_1 + c_2 + c_3 + 1 = 2$

$y'(0) = c_1 - c_3 + 2 = 0$

$y''(0) = c_2 + c_3 + 4 = 2$

which have solutions $c_1 = 3$; $c_2 = -2$; $c_3 = 0$

So the solution to the IVP is $\boxed{y(x) = 3 - 2\mathrm{e}^{x} + \mathrm{e}^{2x}}$

Given an inhomogeneous IVP:

1. Find y_c, a complementary solution to the homogeneous equation.
2. Find y_p, a particular solution to the inhomogeneous equation.
3. Add $y = y_c + y_p$ to get a general solution of the inhomogeneous equation.
4. Apply initial conditions to $y = y_c + y_p$.

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